

StudDFT keywords (short manual, v0.82.1)

Keyword	Options	Synonyms	Designation
Quantum chemical theories			
%method			Required. Electronic-structure method (case-insensitive).
	HF		Hartree-Fock
	MP2		HF reference + MP2 correlation (post-SCF)
	SVWN	SVWN5, LDA, LSDA	Slater + VWN-V LDA (NWChem / PySCF convention)
	SVWN3		Slater + VWN-III RPA (Gaussian 09 convention)
	BLYP		Becke88 + LYP GGA
	BP86	BP	Becke88 + Perdew86 GGA
	BPW91		Becke88 + Perdew-Wang 91 GGA
	PBE		Perdew-Burke-Ernzerhof GGA
	PBE0	PBE1PBE	PBE with 25 % HF exchange
	B3LYP	B3LYP5	B3LYP with VWN-V correlation (ORCA / NWChem)
	B3LYP3		B3LYP with VWN-III correlation (Gaussian convention)
			Note: open-shell calculations automatically use U-prefixed variants (UHF, UMP2, UB3LYP, ...). TPSS is not available as an SCF method but TPSS-specific D3(BJ) and D4(BJ) dispersion parameters are present.

Job type			
%job			Task to run. Default: energy.
	energy	sp, single	Single-point energy (default)
	gradient	grad, force	Analytical gradient at fixed geometry
	opt	optimize	Geometry optimization (internals RFO or Cartesian BFGS, see %opt_coord)
	freq	frequency, hessian	Hessian + harmonic vibrational frequencies
	opt+freq	optfreq	Opt followed by freq at the optimized geometry
	bsse	counterpoise, cp	Boys-Bernardi counterpoise (requires fragment IDs on every geometry line)
	scan	pescan, rigid_scan	Rigid PES scan; requires %gz block and %scan ... *end block
	scanopt	relaxed_scan, scan+opt	Relaxed PES scan: at every grid point a constrained geometry optimization is performed
	ts	opt_ts, optts	Transition-state search via P-RFO (HF / DFT only)
	ts+freq	tsfreq, opt_ts+freq, opttsfreq	TS search followed by freq pass at located TS
Molecule builder			
%charge	<signed integer>		Total molecular charge. Default: 0.
%mult	<positive integer>	%multiplicity	Spin multiplicity $2S+1$ (1 = singlet, 2 = doublet, ...). Default: 1.
%spin	restricted unrestricted	RHF, RKS, R UHF, UKS, U	SCF spin treatment. If not set, determined automatically from %mult.
%geo	[angstrom bohr]	%geometry, *geometry	Start of Cartesian geometry block. Terminated by *end or

			empty line. Atom token: <Z> <Element> <Element><index>.
%gz			Start of Z-matrix geometry block (Angstroms, degrees). Terminated by empty line. Required for %job scan / scanopt.
Basis set			
%basis	<basis set name>		Required. Basis set name (case-insensitive). Built-in: sto-3g, 3-21g, 6-31g, 6-31g(d), 6-31g(d,p), 6-31+g(d), 6-31++g(d,p), 6-311g(d), 6-311g(d,p), 6-311g(2d,2p), 6-311g(3df,2pd), 6-311++g(2d,2p), 6-311++g(3df,2pd), cc-pVTZ, cc-pVQZ, aug-cc-pVDZ, aug-cc-pVTZ, aug-cc-pVQZ, LANL2DZ. MIXED: per-atom assignment via <selector> <basis_name> lines until next %-keyword or empty line.
%spherical	(flag)		Use spherical-harmonic d/f-shells (5d/7f); default is Cartesian (6d/10f).
ECP			
%ecp	<ecp name>		Effective core potential. Built-in: LANL2DZ. MIXED: per-atom assignment via <selector> <ecp_name> lines until next %-keyword or empty line.
%ecp_method	[numerical] analytical	numerical: quadrature, num, 0 analytical: md, mcmurchie, 1	ECP integral evaluation method. Default: numerical (Mura-Knowles radial quadrature). Analytical = McMurchie-Davidson; substantially overhauled in v0.77.0-0.77.1 (now matches g09 to ~1 microHa on AlH3-

			class molecules). For very tight basis primitives ($\alpha > \sim 5000$) prefer numerical.
Electronic integral calculations			
%boys_cook	(flag)	%boys_precise	Use Cook recurrence for Boys function instead of fast tabulated representation. Default: off (table method).
SCF procedure			
%scf_converger	[DIIS] NONE	%converger	Convergence accelerator. Default: DIIS. Internally tests whether value starts with "DIIS"; anything else turns DIIS off.
%scf_maxiter	<positive integer>		Max number of SCF iterations. Default: 150.
%scf_etol	<positive real>		Energy convergence threshold (Hartree). Default: 1e-8.
%scf_ptol	<positive real>		Density-matrix convergence threshold. Default: 1e-6.
%scf_diis_etol	<positive real>	%scf_gtol	DIIS error-norm convergence threshold (v0.76.0). All three (%scf_etol, %scf_ptol, this one) must be satisfied simultaneously. Default: 1e-5.
%scf_min_iter	<positive integer>		Minimum SCF iterations before convergence can be declared (v0.76.0). Default: 5.
%scf_n_converged	<positive integer>	%scf_nconv	Consecutive iterations all three convergence criteria must be satisfied (v0.76.0). Default: 2.
%scf_diis_cond_max	<real > 1>	%scf_diis_condmax	Upper bound on DIIS B-matrix condition number; oldest vector is pruned when exceeded (v0.76.0). Default: 1e10.

%damping	<real in [0,1]>		Density-matrix damping factor. DIIS overrides damping after a few iterations. Default: 0.0.
%level_shift	AUTO OFF <non-negative real>		Virtual-orbital level shift (Hartree). Useful for problematic open-shell / heavy-element cases. Default: AUTO (turning on shift 0.2 if SCF is not converged in 100 steps)
%initial_guess	AUTO SAD CORE	%guess, %scf_guess, HCORE, CORE_H	Initial-guess strategy (v0.76.0). Default: RHF/RKS - CORE, UHF/UKS - AUTO (SAD if a fit basis is loadable, CORE otherwise).
%uks_occupation	AUTO STRICT MOM MIX	%uhf_occupation, %occupation, AUFBAU, MAXOV, MAX_OVERLAP, REMIX	UHF/UKS orbital-occupation strategy (v0.76.0). Default: AUTO. No effect for restricted.
%eri_memory_limit_mb	<positive integer, MB>	%eri_mem_limit_mb, %eri_memory_limit, %eri_mem_limit	ERI-buffer memory cap (v0.76.3); falls back to direct JK when exceeded. Default: 4096.
DFT options			
%grid		%grid_quality	DFT integration-grid quality. Default: 3 (fine, 75 radial). No effect for HF or MP2.
	1		Coarse (25 radial)
	2		Medium (50 radial)
	3		Fine (75 radial; default)
	4		Ultrafine (100 radial)
	5		Extreme (150 radial)
	fine	gfine	Preset 6: Gaussian Fine (75 radial, 302 Lebedev, Euler-Maclaurin + size-adjustment)
	ultrafine	gultrafine	Preset 7: Gaussian UltraFine (99 radial, 590 Lebedev)

	tm	treutler	Preset 8: Treutler-Ahlrichs (75,302) uniform 302-point angular, no pruning (replaces former pyscf alias in v0.77+)
	g09	g09ref	Preset 9: G09 reference (75,302) uniform 302-point angular, Euler-Maclaurin + size-adjustment
%xc_deriv	[analytical] numerical	analytical: anal, exact numerical: num, fd	Analytical or finite-difference XC functional derivatives during SCF and DFT gradient / Hessian. Default: analytical.
Dispersion			
%dispersion	NONE D3	%disp, D3BJ, D3-BJ	Empirical dispersion correction. Default: NONE. D3(BJ) params: HF, BLYP, BP86, BP, BPW91, B3LYP*, PBE, PBE0.
MP2 options			
%frozen_core	-1 0 <positive integer>	%nfrozen	Frozen-core control. -1 (default) = auto-detect noble-gas core of every non-ECP atom; 0 = correlate all; N = freeze lowest N occupied orbitals (per spin for UMP2).
%pump2	[yes] no	on, true, 1, T, <bare> off, false, 0, F	Spin-projected open-shell MP2 (Schlegel-1986 single-annihilation). Default: yes -- PUMP2 single-point diagnostic always printed when UMP2 is active. With %pump2 yes, FD-PUMP2 optimization is also enabled (gradient w.r.t. PUMP2 energy). With %pump2 no, PUMP2 is fully suppressed.

Geometry optimization			
%geom_maxiter	<positive integer>	%opt_maxiter, %opt_max_iter, %geom_max_iter	Max optimization steps. Default: 50.
%grad_tol	<positive real>		Maximum-component gradient convergence threshold (Ha/Bohr). Default: 4.5e-4.
%grad_method	[analytical] semianalytical	analytical: full, 1 semianalytical: semi, fd, 0	Gradient method. PUMP2 gradient is always numerical (h = 1e-3 Bohr).
%opt_coord	[internal] cart	%opt_coordinates, %opt_coors internal: internals, redundant, rfo, int cart: cartesian, bfgs, cart_bfgs, cartesian_bfgs, xyz	Optimizer coordinate system (v0.75.0). Default: internal (redundant- internals RFO). Cartesian path does NOT support constraints or PCM.
Geometry constraints (partial-constraint optimization)			
%constrain			Start a constraint block (v0.70+). One declaration per line, terminated by *end or empty line. Compatible with %job opt / opt+freq, internal- coordinate optimizer only.
	B <i> <j>		Freeze bond between atoms i and j
	A <i> <j> <k>		Freeze angle i-j-k (vertex at j)
	D <i> <j> <k> <l>		Freeze dihedral i-j-k-l
PES scan (rigid / relaxed)			
%scan			Start a sweep-variable block (v0.69+). Required by %job scan and %job scanopt. One sweep variable per line, terminated by *end or empty line. Requires a %gz Z-matrix geometry.
	<kind> <row> [from] <val_from> [to] <val_to> [step] <val_step>		<kind> = R (bond) A (angle) D (dihedral); <row> = 1-based Z-matrix row index. Keywords from/to/step are optional.

Hessian / Frequencies			
%hess_method	[analytical] numerical	analytical: full, 1 numerical: num, fd, 0	Hessian computation method. Default: analytical (HF, DFT, ECP). Numerical = finite difference on analytical gradient (h = 5e-3 Bohr).
TS options (transition-state search)			
%read_hess	<filename>	%read_hessian	.hess file to load before TS search (v0.72.0). Reader verifies geometry, method, basis, ECP, charge, multiplicity, spin, format version. If omitted, the TS driver computes the Hessian itself at the input geometry.
%ts_mode	<positive integer>	%tsmode	1-based mode index to follow on the first P-RFO step (v0.72.0). After step 1, optimizer switches to maximum-overlap mode tracking. Default: 1 (lowest mode).
PCM (implicit solvent, v0.73.0+)			
%pcm	none cpcm iefpcm cosmo	cpcm: c-pcm, c_pcm iefpcm: ief-pcm, ief_pcm, pcm none: off, no, false, 0, F	Continuum solvation model. Default: none. %solvent or %eps auto- activate CPCM if no %pcm given.
%solvent	<name>		Named solvent (resolved to tabulated permittivity; overrideable by %eps). Recognized: water (h2o), methanol (meoh), ethanol (etoh), acetone, acetonitrile (mecn), dmso, dmf, thf, chcl3 (chloroform), ccl4, dichloromethane (dcm, ch2cl2), toluene, benzene, cyclohexane, hexane (n- hexane), diethylether (et2o, ether), ethyl_ acetate (etoac), ammonia (nh3), vacuum (none).

%eps	<real >= 1.0>	%epsilon, %pcm_eps	Direct override of dielectric permittivity. Vacuum = 1.0.
%pcm_lebedev	<positive integer>		Per-atom Lebedev quadrature order on cavity surface. Default: 110. Allowed: 6, 14, 26, 38, 50, 74, 86, 110, 146, 170, 194, 230, 266, 302, 350, 434, 590.
%pcm_scaling	<positive real>		Scaling factor on Bondi vdW radii for atomic-sphere cavity. Default: 1.20 (standard COSMO/CPCM).
%pcm_switch	<positive real, Bohr>		Width of cubic-taper switching function on cavity-tessera occupancy. Default: 0.20 Bohr (typical 0.1 - 0.4).
%pcm_grad_method	[analytical] numerical	%pcm_gradient_method analytical: ana, an, 1, true, yes, on numerical: num, fd, 0, false, no, off	PCM gradient method (v0.74.x). Default: analytical (fast). Numerical = central-difference FD, ~10-30x slower; agrees with analytical to ~1e-5 a.u./bohr.
Output control			
%print_mo	(flag)		Print converged MO coefficient matrix at end of SCF.
%print_time	(flag)	%print_timing	Print detailed timing breakdown per timer counter at end of run.
%print_grad	(flag)	%print_gradient	Print per-atom Cartesian gradient at every optimization step (format: <Element><index> gX gY gZ).
%export_mo	(flag)		Save converged MO coefficient matrix in Molden format to the external file <base>_mo.molden

Comments			
# or !			Comment markers at start of line. Whole line is ignored. Inline comments are NOT supported -- everything after the keyword on the same line is parsed as the value.